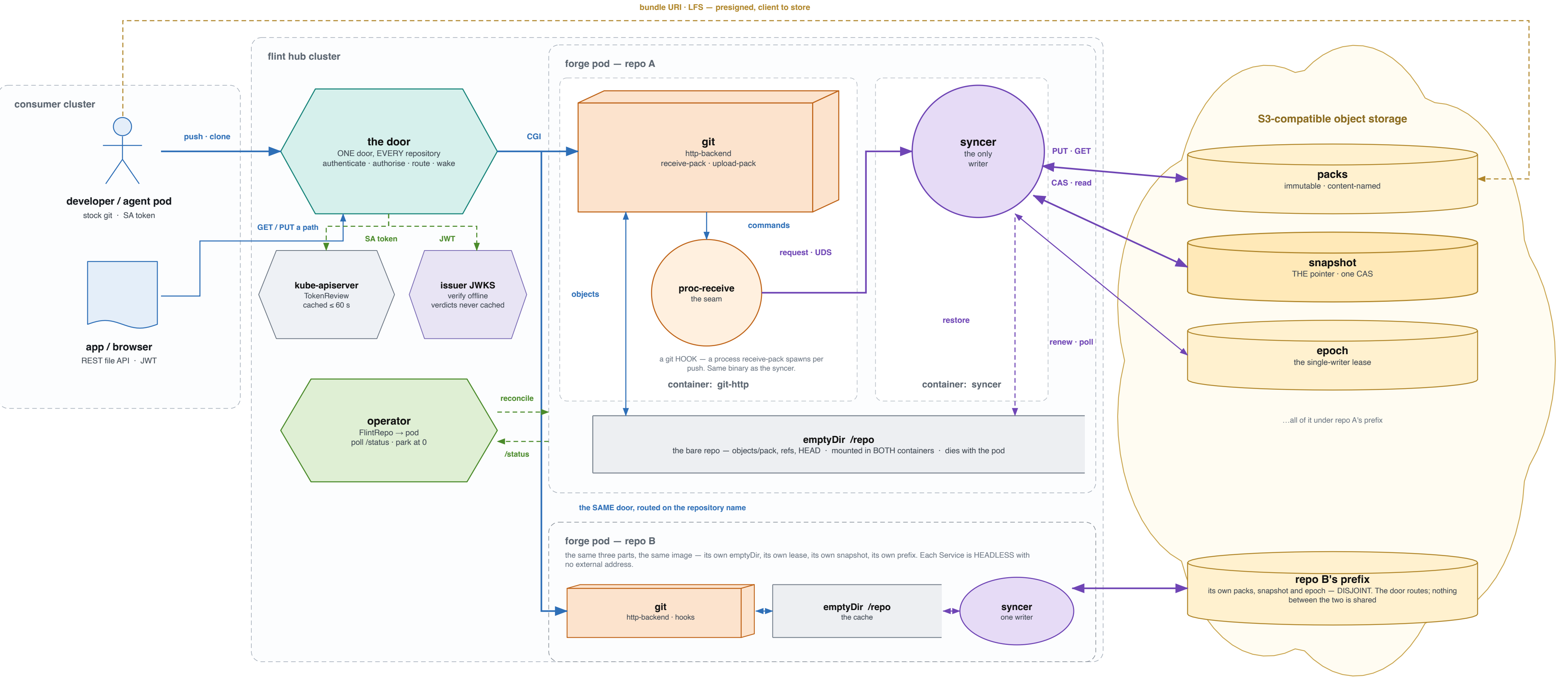


Data flow

Stock git serves every byte. ONE stateless door fronts every repository; one syncer beside each is the only process that can write S3.



ONE DOOR, MANY REPOSITORIES — and this is exactly where forge and flint-lite differ. The door holds no repository state: it verifies the token, resolves the repository named in the URL to its FlintRepo, routes to that repository's pod and wakes it if it is parked — so a thousand repositories are ONE endpoint, one DNS name and one credential shape to every client. A lite hub cannot be fronted that way, because the hub IS the tree: one process, one prefix, dialled directly at the address its own Service advertises. A second lite volume is a second pod, a second Service and a second mount at the consumer. And the door is not only a router: where the operator renders the NetworkPolicy, it is the ONLY peer admitted to a repository pod's git port — which is what makes X-Remote-User mean anything. Where that policy is NOT rendered, reaching the port IS the authorisation — and the operator raises PrincipalsUnenforced against that repository rather than leaving it unsaid. Either way a client that bypasses the door is already outside the trust model, so the door is a COMPLETE oracle of who pushed and when — which is what the idle ladder counts. A lite hub has no such place.

TWO CREDENTIAL KINDS, SERVED AT ONCE. The door routes on the token's iss: a pod's ServiceAccount token goes to TokenReview at the apiserver, an issuer-minted JWT to offline signature verification against the cached JWKS. A JWT's sub — a person — becomes the principal for spec.consumers, for the branch policy, and as the commit author.

AND THE DOOR REFUSES TO START if the cluster's ServiceAccount issuer and the configured JWT issuer are the same string: every pod token would then route to the offline verifier, which cannot know a pod was deleted.

ONE CAS ON THE SNAPSHOT is the transaction boundary. Told “ok” means that object moved; a 412 on it is a fence, not a retry.

→ data plane — every byte of a clone or push → durable path — the only writer of the bucket → control plane — never reads an object → bypass — the pod sees a few hundred bytes

What the abbreviations mean

CAS compare-and-swap — a write that lands only if the object is still the version you read	ETag / If-Match S3's precondition: the ETag you read, offered back as the terms of the write	fence a lost CAS. Another server holds this repository, so this one stops READING too, and exits	CGI Common Gateway Interface — how the runner invokes git http-backend, once per request
UDS Unix domain socket — the hook-to-syncer channel, inside the pod, never over a network	proc-receive the git hook that lets the server, not git, decide each ref update	pre-receive the git hook that applies the branch policy, and can refuse the whole push	emptyDir a Kubernetes volume that lives and dies with the POD — not with a container
SA token a Kubernetes ServiceAccount token, projected into the pod and rotated by the kubelet	TokenReview the Kubernetes API that says whether a ServiceAccount token is still valid	JWT JSON Web Token — a signed bearer token, verified at the door with no round trip	JWKS JSON Web Key Set — the issuer's signing keys. The KEYS are cached, the verdicts are not
iss / sub / exp a JWT's issuer, its subject — the principal — and its expiry	LFS Git Large File Storage — big blobs kept beside the repository, moved by presigned URL	bundle URI git's own opt-in: the server names a pack the client fetches straight from S3	FlintRepo the custom resource that one forge repository is declared as